DT124A - Embedded Systems

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1 Introduction

This is the final lab of the course and is intended to showcase what you have learned throughout, with some additional integration of concepts.

1.1 Purpose

The purpose of this lab is:

- Design and implement an embedded system
- Document said embedded system

1.2 Equipment

- Laptop computer (students' own) with Microchip Studio IDE (or equivalent IDE)
- Arduino Uno (or similar microcontroller) and, possibly, AVR programmer
- Red and blue LEDs
- Resistors (1 x 220Ω), and if needed others for LCD, see separate documentation.
- 16x2 LCD display.
- Sensors.
- Buttons.

Preparation

In order to do this lab, you need to:

1. Read the section about theory.
2. Prepare the lab.

2 The Project

The purpose of the project is to make an embedded system prototype that has several different sensors, at least one button to switch between sensors, and some interface to show the read value from the sensor. The finished project should be done as an embedded system project that is ready to hand off to an end user, therefore also requiring a basic user manual.

2.1 Requirements for the project:

These are the requirements for the project based on the grade you wish to achieve. I will count a potentiometer as a sensor.

2.1.1 Grade 3:

- A working Arduino project prototype that fulfills the following requirements:
  1. At least two sensors.
  2. A button to switch between the sensors.
  3. Output that is in some way human-readable.
  4. Error indicator: Two LEDs (blue and red) where blue indicates normal function of the prototype and red indicates an error. *Students must choose what is normal function and when there is an error based on their knowledge of embedded systems*
- A report on this prototype, see the section on the report.
- A user manual for the end user of this prototype
- Clear commented code
- A in-person presentation of your work during the lab time, where you need to show an almost full working project.
- A short recording where you explain the code.

2.1.2 Grade 4:

- A working Arduino project prototype that fulfills the following requirements:
  1. At least three sensors.
  2. A button to switch between the sensors readings.
  3. Output from the sensors to be shown on the LCD screen, human-readable.
  4. Error indicator: Two LEDs (blue and red) where blue indicates normal function of the prototype and red indicates an error. *Students must choose what is normal function and when there is an error based on their knowledge of embedded systems, this must be a reasonable error*
- A report on this prototype, see the section on the report.
- A user manual for the end user of this prototype
- Clear commented, well written, code:
  1. Interrupts are kept short, no logic only changes to what needs to be changed
  2. Constants are in CAPITAL LETTERS
  3. Variable name reflects usage
  4. Function name reflects usage
  5. See the documentation lecture slides for tips.
  6. The c-file should include a comment header telling who made it, last changes, what it does etc.
- A in-person presentation of your work during the lab time, where you need to show an almost full working project.
- A short recording where you explain the code.

2.1.3 Grade 5:

- A working Arduino project prototype that fulfills the following requirements:
  1. At least three sensors.
  2. 3 buttons to choose which sensor is being use, this is the replacement for the one button system from before where you switched between sensors and now you choose directly which sensor.
  3. Output from the sensors to be shown on the LCD screen, human-readable. Two rows; Row mode (i.e Voltmeter) and the other with measurement and the unit (i.e 5mV)
  4. Error indicator: Two LEDs (blue and red) where blue indicates normal function of the prototype and red indicates an error. *Students must choose what is normal function and when there is an error based on their knowledge of embedded systems, this must be a reasonable error*
  5. Separate adc.c and adc.h where the handling of the ADC is done.
  6. Student initiative: Some thing that is not in the requirements but you think could fit for this embedded system prototype.
- A report on this prototype, see the section on the report.
- A user manual for the end user of this prototype
- Clear commented, well written, code:
  1. Interrupts are kept short, no logic only changes to what needs to be changed
  2. Constants are in CAPITAL LETTERS
  3. Variable name reflects usage
  4. Function name reflects usage
  5. See the documentation lecture slides for tips.
  6. The c-file should include a comment header telling who made it, last changes, what it does etc.
- A in-person presentation of your work during the lab time, where you need to show an almost full working project.
- A short recording where you explain the code.

3 Report

- Introduction
  1. Aim
  2. Explanation of the different sensors
- Method and Equipment
  1. Design
  2. Hardware component list
  3. Software design
  4. Optimization
  5. Tests
- Result

1. Results from the tests
2. Flowcharts or other diagrams to show how the prototype works
3. User manual, see separate section

- Discussion
- Conclusion
- References

4 User manual

The user manual should have the following headings:

- Introduction: This section should provide an overview of the purpose of the manual and the embedded system it covers.
- System Overview: This section should provide an overview of the embedded system, including its architecture, components, and interfaces.
- Hardware Requirements: This section should list the hardware requirements for the embedded system, including any specific components that are necessary for proper operation.
- Software Requirements: This section should list the software requirements for the embedded system, including any specific operating systems, drivers, or applications that are necessary for proper operation.
- Installation: This section should provide step-by-step instructions for installing the embedded system, including any necessary software and hardware.
- Usage: This section should provide instructions for using the embedded system, including how to interact with the system and any specific features or capabilities.
- Troubleshooting: This section should provide guidance for troubleshooting common issues that may arise when using the embedded system.